Assignment #5

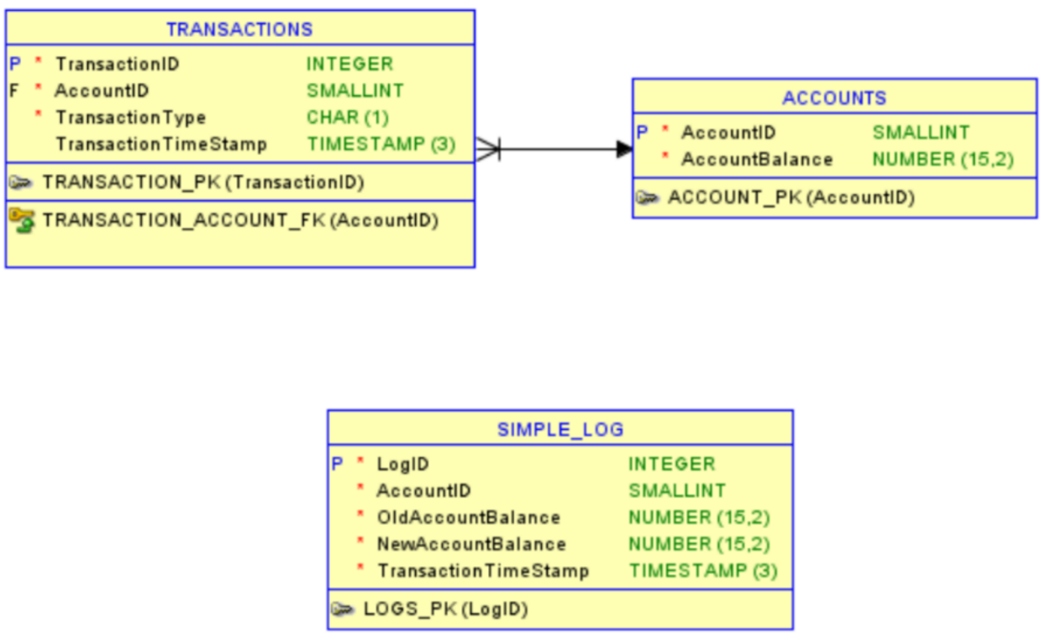
Course: ISA 345

Instructor: Dr. Maliha Alam

Points:100

Submission instructions: this assignment is to be done individually. The answers to the first question must be in a single document, *e.g.*, a *doc* or a *pdf* file.

Question 1: Reconsider the accounting-system example studied in class during Lecture 23, which is shown below:



Before solving the following questions, make sure you download the file *A5Q2.sql* available on Canvas and run all the statements in it to create the required tables, data, and trigger.

a) Write a procedure called TRANSFER, which has the following arguments: *AccountIDFrom* NUMBER, *AccountIDTo* NUMBER, *TransferAmount* NUMBER. The TRANSFER procedure consists of 4 operations:
    1. A new transaction of type 'D' (debit) is stored in the TRANSACTIONS table. The *TransactionID* of this transaction is equal to the previous

TransactionID plus one. The *AccountID* of this transaction is *AccountIDFrom;* **[5 points]**

2. Update the balance of the account with *AccountID* equal to *AccountIDFrom.* The new balance is equal to the old balance minus *TransferAmount*; **[5 points]**
3. A new transaction of type 'C' (credit) is stored in the TRANSACTIONS table. The *TransactionID* of this transaction is equal to the previous *TransactionID* plus one. The *AccountID* of this transaction is *AccountIDTo;* **[5 points]**
4. Update the balance of the account with *AccountID* equal to *AccountIDTo.* The new balance is equal to the old balance plus *TransferAmount.* **[5 points]**

b) Show that the TRANSFER procedure works. In particular, simulate a scenario where a person with *AccountID* equal to 1 transfers $10 to a person with *AccountID* equal to 2. To do so, show the resulting tables after running each one of the following commands: **[5 points]**

```
SELECT * FROM ACCOUNTS ;
SELECT * FROM TRANSACTIONS;
CALL  TRANSFER (1, 2, 10);
SELECT * FROM ACCOUNTS;
SELECT * FROM TRANSACTIONS;
```

c) Suppose that a fraud-detection system will use data from SIMPLE_LOG to try to detect and prevent fraudulent transactions. Due to an internal miscommunication, the fraud-detection system requires data about the transferred amount (*NewAccountBalance – OldAccountBalance*), as opposed to data about new balance (*NewAccountBalance*) or old balance (*OldAccountBalance*). Instead of redeveloping the whole fraud-detection system or a new database, you realize that the above problem can be fixed by simply creating a VIEW on top of SIMPLE_LOG, called *ADAPTED_LOG*, that returns the four attributes required by the fraud-detection system, namely *LogId*, *AccountId*, *TransferredAmount*, and *TransactionTimeStamp*. Report the SQL code to create *ADAPTED_LOG.* **[10 points]**

d) Show that *ADAPTED_LOG* works as desired by showing the resulting tables after running each one of the following commands: **[5 points]**

```
SELECT * FROM ADAPTED_LOG;
CALL  TRANSFER (3, 4, 50);
SELECT * FROM ADAPTED_LOG;
```

Recall that there is a trigger inside *A5Q2.sql* that is executed whenever there is an update of the attribute *AccountBalance* on the table ACCOUNTS.